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Day Month Year

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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter ??: Not started. Will write last/almost last
- Chapter ??: Some ideas for content, but no words written yet.
- Chapter ??: Approaching first draft. Lots of work is still required to make clear that terms are properly defined.
- Chapter 2: Vague idea for structure, but almost no words have been written.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter ??: Passed first drafting. Awaiting Ken and Bryan's comments.
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- Appendix ?? : Mostly written
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Declaration

Acknowledgements

Chapter 1

Abstract

Chapter 2

High Power Laser Prototype for LIGO

2.1 Introduction

LIGO requires high input power, nominally 125 W, to decrease the shot noise which limits its sensitivity at high frequencies ($> 100\text{Hz}$). The ‘High Power Oscillator’ (HPO) was designed to provide this power. The HPO was a ring cavity containing laser crystals and would be seeded by a 35W laser. This design went through extensive testing and refinement at the AEI [1], and was installed as part of the LIGO ‘Pre-Stabilised-Laser’ (PSL). However, due to jitter noise coupling to the differential arm motion signal (DARM) from the water cooling of the the crystals within the HPO (as well as one version being destroyed by accident) [2], the HPO was bypassed, and LIGO operated using the 35 W laser.

The use of a single pass amplifier was investigated as a replacement of the HPO, and for third observation run a Neovan-4S deployed to step the PSL power up from 35 W to 70 W [3]. The next stage of development was a complete redesign of the PSL based around using two upgraded versions of the Neovan-4S, the Neovan-4SHP. This amplifier showed promising results [4, 5], so a prototype PSL was constructed and tested at LIGO Livingston. This design will be installed as the new PSL for the start of O4 [2].

2.2 Properties of the LIGO PSL

A laser has five properties: intensity, frequency, beam shape, pointing and polarisation. The PSL must provide a beam which satisfies the requirements for each of these properties [6]. Along with these basic properties of a laser beam, the PSL must also be able to be operated without interruption for timescales of the order 1 year, have remote diagnostic tools, and have a remote interface so that operators can control and monitor the status of the laser.

2.2.1 Intensity

The PSL should deliver 125 W to the interferometer. The intensity fluctuations should be lower than the noise generated by radiation pressure noise causing motion of the test masses. The intensity noise should be limited by shot noise around 100 Hz where radiation pressure noise has dropped off. Above the bandwidth of the cavities, the intensity noise requirement can be relaxed as the interferometer becomes less sensitive to intensity noise.

There are several ways of stabilising a laser. Modulation of one of the amplifiers pump diodes and intensity actuation by way of an acoustic optical modulator (AOM) were investigated during the prototyping of the new PSL design. By modulating the one of the pump light diodes for one of the amplifiers, we are able to reduce the amount of transmissive optics in the high power path of the beam by one. This is an attractive idea since the layouts are simpler to set-up and have less high power components. The AOM method is the one currently in use for intensity stabilisation at LIGO.

2.2.2 Frequency

Ultimately, LIGO uses the arm cavity as a frequency reference since $\delta f/f = \delta l/l$, however the PSL must provide a suitably low frequency noise to the interferometer so that it can acquire lock. The PSL uses an ultra low expansion fused silica reference cavity in a Pound-Drever-Hall (PDH) locking scheme to provide this frequency stabilisation [7].

2.2.3 Beam Shape

The PSL should produce a high quality TM00 beam. Higher order modes do not enter the interferometer, so while higher order modes at this point of the interferometer do not couple directly to the gravitational wave, they do result in a drop in power. Additionally, reflected light from the interferometer is used to derive control signals, thus unwanted light due to higher order modes from the PSL will contribute to the noise of these signals.

2.2.4 Pointing

The pointing noise witnessed at LHO is one of the reasons why single pass amplifiers were chosen as the replacement of the HPO. Pointing noise manifests as intensity noise since a change in beam direction can be modelled as the introduction of higher order modes. These new amplifiers are cooled in such a way that the jitter on the PSL beam is reduced.

2.2.5 Polarisation

Highly linear polarised light is essential to the operation of LIGO, since the optical coatings are all optimised for a specific polarisation.

2.2.6 Other Aspects of the PSL

It is useful to diagnose the PSL remotely, so a diagnostic breadboard (DBB) [8] capable of making pointing, intensity and frequency noise measurements, as well as mode scans, was integrated into the design of the PSL. Remote operation and monitoring is also required, so slow control is provided by a Beckhoff system.

2.3 Set-up and Layout

2.3.1 Concept

The new design of the PSL is based on using an NPRO with two single pass amplifiers in series, followed by a cavity to provide 125W of laser light with high spatial purity and low noise. Currently, the PSL is seeded by a 35W laser which is followed by one amplifier [3, 2]. Results from the high performance version of this amplifier were

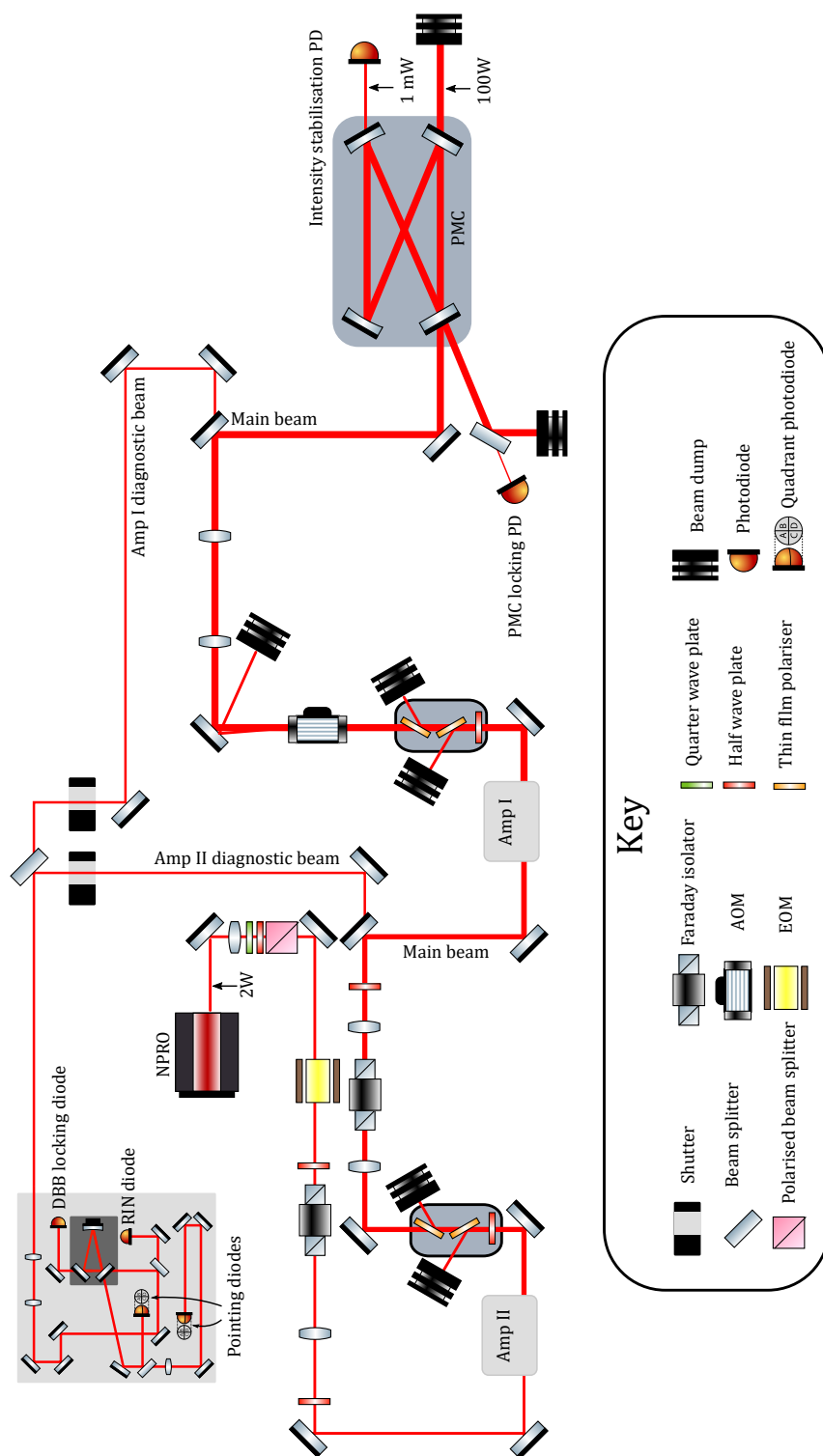


Figure 2.1: Sketch of PSL

promising [4], leading to a proposal to construct the PSL out of two of these in series. A similar set up has been constructed [5]. The cavity can be considered as a reference beam parameter and a hand-off point between the PSL and the rest of the interferometer. This cavity is known as the PMC. The PMC has been redesigned [9]; it is no longer in a vacuum enclosure and it used new mirror mounts. The PSL is housed in an environmentally controlled room so that it is not affected by temperature and humidity changes.

2.3.2 NPRO

The first element of the PSL is a non-planar ring oscillator laser [10, 11, 12]. This is a single frequency laser constructed from one Nd:YAG crystal which acts as both the gain medium and the cavity, thus it produces a laser with high amounts of frequency stability. The NPRO used is a Mephisto laser, built by Coherent, and it can produce to 2W of 1065 nm light.

2.3.3 Neovan 4S-HP solid-state Amplifier

The Neovan 4S-HP is a single pass, solid state amplifier manufactured by Neolase. It is constructed from four Nd : YVO₄ crystals which are pumped by fibre coupled laser diodes. It should be seeded by s-polarised light at 1064 nm. It is cooled via a plate with water (or other coolants) flowing through it. This keeps jitter noise low as flowing water is not directly in contact with the gain medium.

2.3.4 Diagnostic Bread Board

The diagnostic bread board (DBB) is able to make intensity noise, frequency noise, beam pointing noise and higher order mode content measurement [13, 8]. The breadboard consists of a cavity, motorised lenses, an automatic alignment system and several photodiodes. It is integrated into the PSL via several pick-off points and can be used to monitor the status of the NPRO, Amp I and Amp II.

2.3.5 PMC

The pre mode cleaner (PMC) is used as the hand-off point between the PSL and the rest of the interferometer. Light transmitted by this cavity is used as the input beam for the interferometer. The PMC is a bow tie cavity with an FSR of 9 MHz, thus it passively filters RF intensity noise. Control side bands are provided after the PMC but before the IMC, meaning that the IMC cannot be used to provide this stabilisation. The PMC is locked to the fundamental mode via a PDH lock, so the beam that is delivered to the rest of the interferometer has a low amount of higher order mode content. This is important as the amplifiers upstream of the PMC produce slightly elliptical beams. The PMC can be considered as a reference point for mode matching, with the intention being that one can work on the PSL and have a set of reference beam parameters to work towards.

2.4 Set up Procedure for the Neovan 4S-HP

To ensure that a large portion of the light exiting the amplifier is in the fundamental mode, the alignment, mode matching and polarisation of the input beam must be

correctly set. The optimal alignment and mode matching will correspond to the maximum power in the TM00 mode that can be obtained for a given pump diode current, cooling, and input power. This is complicated by thermal lensing effects and by the nature of working with high power beams.

2.4.1 Balancing all variables to maximise power through the PMC

There are six variables that will effect the amount of power that is transmitted through the PMC: the flow rate through the amplifiers heads; the input alignment to each amplifier; the mode matching between to each amplifier; the mode matching to the PMC; the alignment to the PMC; the pump diodes current and the polarisation of the light. These are not all independent of each other. For instance, changing the flow rates will change the thermal lensing in the amplifier. This will change the optimal mode matching and alignment to the amplifier. Changing the the pump diode current effects the thermal lensing as well, meaning increasing it doesn't always lead to an improvement in TM00 power.

The flow rates to the amplifiers should be decided before spending time optimising the alignment and mode matching. Then the alignment, mode matching and pump diodes should be tweaked in tandem. This is an iterative process, and each time the thermal lensing is changed, in order to properly measure the higher order mode content of the beam the DBB mode matching should be updated. This process takes a long time, as each time a change in either the alignment, mode matching or diode current is made, the DBB mode matching needs fixing. Due to the range of tunability on the DBB mode matching lenses, this took a long time.

2.4.2 Flow Rates and Cooling

Water flow example. The first thing that should be decided is how the cooling will be done. The flow rates of the coolant to the amplifier affects the the amount of thermal lensing. A balance must be struck between the amount of cooling - more is better - and the amount of jitter that can be potentially introduced to the beam via coolant moving through pipes. Due to the jitter noise experience at the Hanford detector that was due to the water cooling in the PSL, the flow rates will be set as low as possible.

The cooling scheme that will be used is shown in Figure 2.2. Flow meters which monitor the flow rate of the coolant will be placed on the return side of the system rather than the supply side. By having the flow meters on the return side, they will measure a drop in flow if there is a leak, allowing for the water supply to be cut.

To reduce the amount of thermal lensing in the gain medium of the amplifier, the amplifier is cooled. The thermal lensing in the Nd : YVO₄ is non-circular, and since they are cooled from below there is a significant temperature gradient across the crystals. From a beam quality point of view, it is better to have more cooling as with more cooling, there is less thermal lensing.

A different cooling layout was used to test what the appropriate flow rates for the amplifiers and diode boxes could be. The diode box and amplifier head were connected in parallel, and each amplifier was on a separate cooling loop. This has the advantage of being able to change the flow rate through each amplifier independently, meaning that the effect of flow rate on the beam quality can be

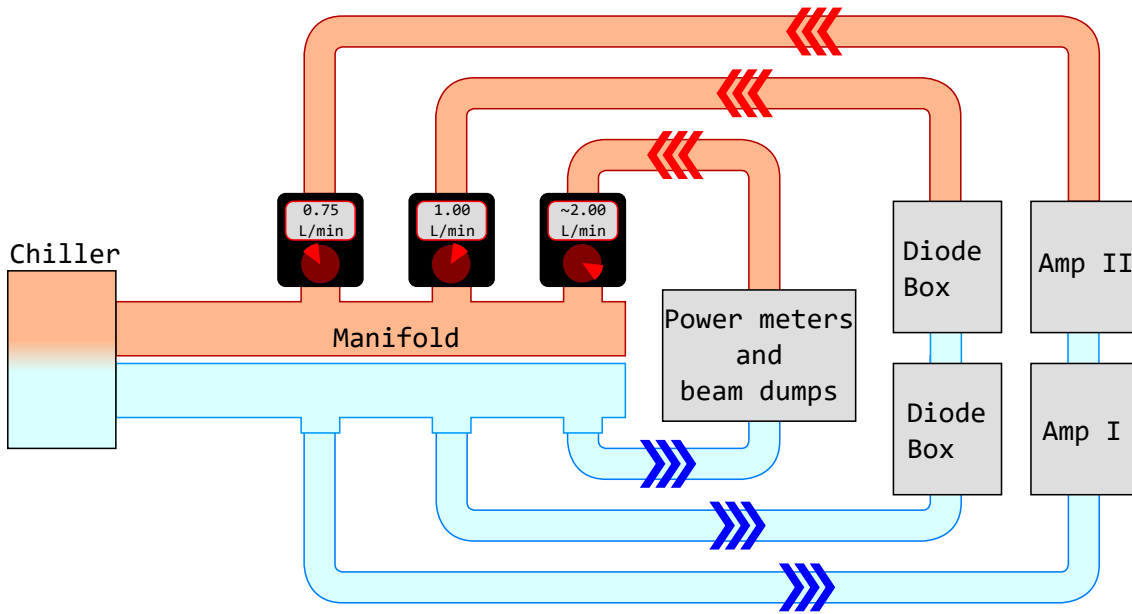


Figure 2.2: Cartoon of the cooling layout for the PSL. The amplifiers are on a separate loop to the diode boxes so that the flow rates for these can be reduced as much as possible.

studied for one amplifier. labbok post

The flow rate to the first amplifier was kept high to get as high a quality beam as possible entering the second amplifier. The HOM content for this beam was 2.8%. The second amplifier and diode box flow rate was decreased. The temperature of the diode head was monitored. The higher order mode content and was measured with the DBB and the power was measured with a high-power power meter.

The results of this are shown in Figure 2.3. As the flow rate was decreased, the temperature of the heads increased. At 22.3°C the higher order mode content was $\sim 4.5\%$ and the amount of power in the fundamental mode was $\sim 127\text{ W}$. It was decided that to keep the amount of power in the fundamental mode high, the diode heads would be operated such that their temperature did not rise above $\sim 27^\circ\text{C}$.

It was decided that the flow rate for the amplifier heads would be set to 0.75 L min^{-1} , as this corresponds to less than 5% HOM and $\sim 125\text{ W}$ of power in the TM00 mode. If a lower flow rate was to be used, the temperature of the amplifiers heads would increase such that the power in the TM00 would fall below 125W. The flow rate to the diode boxes was set to 1 L min^{-1} ; this is as low as possible. If it is much lower then there is a run away heating up effect.

2.4.3 Alignment Procedure

lab book post. First, the amplifier should be switched off. This protects it against damage as well as preventing the possibility of unknown beams with several tens of Watts of power being in the lab. The beam path between the NPRO and the

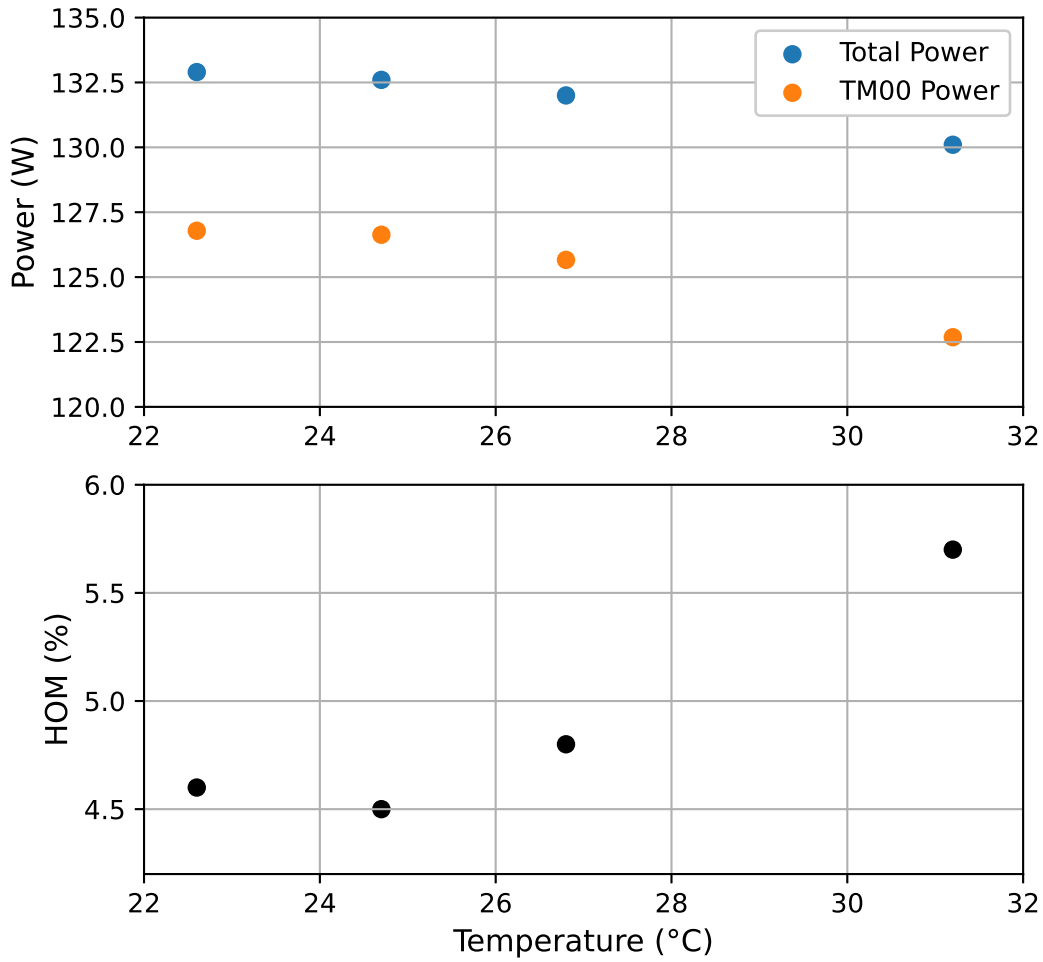


Figure 2.3: As the amplifier heads temperature increases, so does the higher order mode content. The power in the TM00 mode decreases as the temperature decreases due to two effects: less overall power and more higher order modes.

exit of the first amplifier were inspected to make sure the beam path was clear and terminated with dumps.

Next, the NPRO was set to produce 1.3W of light. This power of the light was measured on the input side of the amplifier. Using M03 and M04, the beam was roughly aligned so that the spot would hit the centre of the amplifier's input aperture. The power meter was placed on the exit side of the amplifier, and the input alignment was adjusted until the light exiting the amplifier was the same as the input power. Once this is done, one knows that the beam will be roughly aligned and some amplification will occur when the pump diodes are turned on. Once this was done, the high power attenuator immediately after the amplifier was set to dump all the light. The initial mode matching was based on the manufactures specifications, and the input light was set to s-polarisation.

The pump diode currents were increased until the amplifier 'switched on'. This was at 2 A of current through each of the pump diodes. The amplifier starts amplifying the input light when there is sufficient pump light to cause enough population inversion that stimulated emission occurs.

At this threshold, the output power was maximised by optimising the alignment. Then, the mode-matching was adjusted, in tandem with correcting for misalignments caused by non-perfect centring of the beam on the mode matching lens, L02, and non-perfect co-alignment of the rail this lens is mounted on to the optical axis.

A smaller, faster power meter was used to monitor the power transmitted through the high power attenuator. Care must be taken to ensure the amount of light on this meter does not exceed its damage threshold. Compared to the high-power meter, the low-power meter has less drift and responds to changes in power faster. This gives quicker feedback to the aligner, allowing for easier alignment.

A Wincam was set up to monitor the beam profile. If the amplifier is significantly misaligned and mode mismatched, the beam profile will become distorted. Large misalignment will cause the beam profile to look lopsided, and smaller misalignments will cause the beam profile to have shoulders.

Once the the amplifier has been optimised for a low pump diode current, the pump diode current should be incremented in small steps of roughly 1-2A. Each time this is done, the thermal lensing will increase. Due to the non-uniform way in which the heating occurs, the input alignment needs to be re-optimised each time as does the input mode-matching. The idea is that by going in small steps of diode current, the optimal alignment can be safely achieved. The amplifier should be operated at high diode currents, $\approx 9\text{A}$, so it is not worth spending a huge amount of time perfecting the input alignment and mode-matching at these currents. Its sufficient to be within a 1% of optimal.

Using mode scans from the DBB, one can determine when the alignment to the amplifier is optimal. Since the beam from the amplifier will have a few percent of higher order modes due to thermal lensing, optimising the mode matching from the amplifier to the DBB is difficult. When optimising the pump diode currents, the thermal lensing changes, so the mode matching to the DBB needs to be corrected

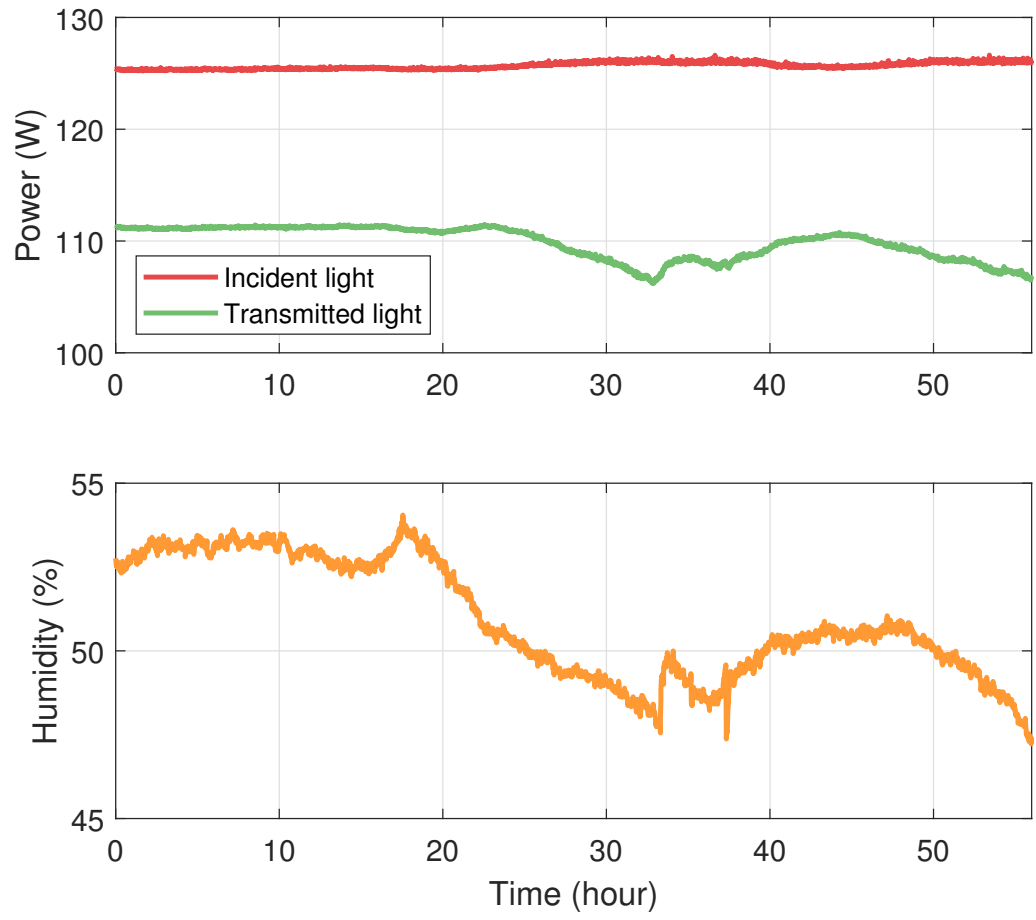


Figure 2.4: Starting on 14th November 2019

for this.

2.5 Output Power

Amplifier	Diodes 1 & 2 Current (A)	Diodes 3 & 4 Current (A)	Output Power (W)	HOM content (%)
Amp II	10.4	9.3	67	4.8%
Amp I	9.5	9.2	137	7.5 ??%

Table 2.1: Amp II: settings on 11/11/2019 post. Amp I: settings on 12/11/2019 post

2.6 Intensity Stabilisation

Read this! The intensity noise requirement for the PSL is found in [6]. At low frequencies, the requirement is driven by the need for the intensity noise to be less than the radiation pressure noise induced by the differences in the arm powers. At 10 Hz, the laser must be stabilised to a level of $\sim 1 \times 10^{-9}$. Above 10 Hz, the requirement on the intensity noise is relaxed as there is passive filtering of the light by the PRC. The light is passively stabilised by the PMC to provide the required RF intensity noise.

The PSL must provide intensity stabilised light to the interferometer. The intensity stabilisation of the laser is done in two stages. Before the the IMC is locked, the laser is stabilised by the inner loop to a level required to the lock the IMC. The light used to measure the intensity noise is a small pick-off beam that is transmitted by the PMC. There is ~ 1 mW of light in this beam, so the laser cannot be stabilised beyond $\sim 2 \times 10^{-8} \sqrt{\text{Hz}}^{-1/2}$ with this beam. Once the IMC is locked, a pick-off from the beam entering the interferometer is used to measure the intensity noise of the laser. This beam has enough power to reach the $\sim 1 \times 10^{-9}$ shot noise requirement. As the inner-loop is in-air and is based on rigidly mounted optics, it is unable to remove noise associated with vibrating mounts.

2.6.1 Free Running Noise of the TnT Prototype PSL

The power noise of the NPRO was measured using the DBB RIN photodiode (Figure 2.7). The TnT NPRO displays a typical noise level that would be expected for this type of laser [12]. This was recorded via CDS, so the measurements near 10 kHz are

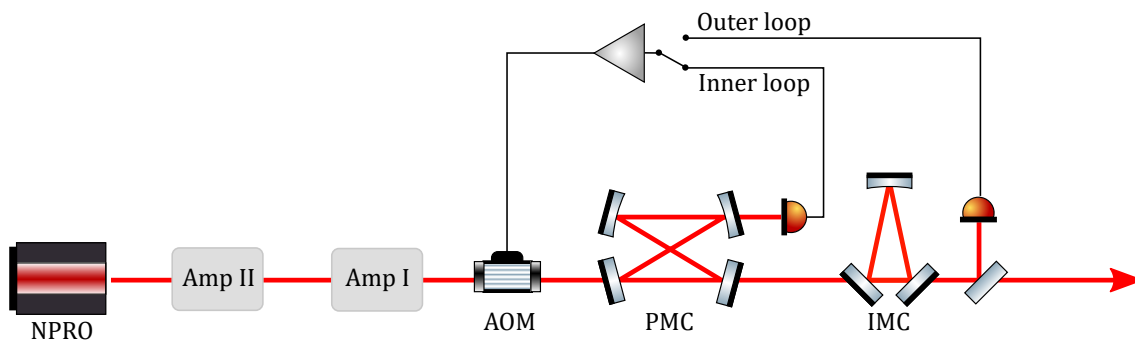


Figure 2.5: ISS concept

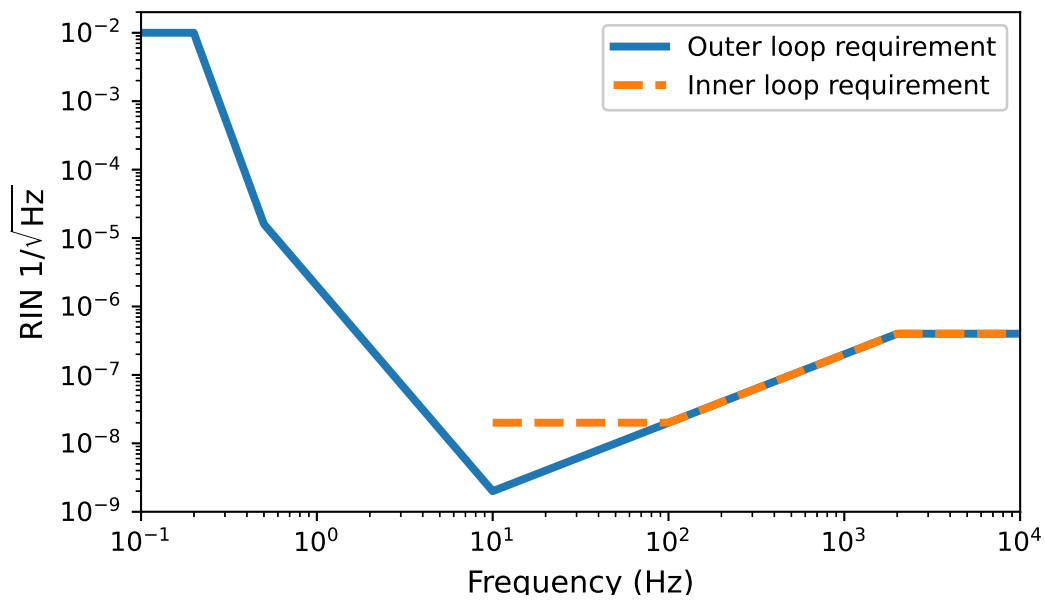


Figure 2.6: ISS concept

effects by the anti-aliasing filter. The noise between 100Hz and 1kHz has features associated with the motion of the table and the mirror mounts.

The power noise of Amp II and Amp I were measured using the DBB photodiode, however the noise was measured via a spectrum analyser rather than CDS, allowing for measurements at higher frequencies (Figure 2.7). The noise of these amplifiers is at a similar level to the NPRO. When the amplifiers are saturated, they produce coherent radiation. This means that if the seed laser changes in power, the amplifiers power will change along with it. The amplifiers are linear for these small power fluctuations, and so the noise *should* be dominated by NPRO noise. However, additional noise is introduced. As there are additional components in the beam path for amp I and amp II compared to the NPRO, there are more lines associated with them between 100Hz and 10kHz. However, the noise of the amplifiers is covered in a forest of lines that are spaced 4Hz, starting from 4Hz and continuing up to high frequency. These lines were found to originate in the Beckhoff slow control computer which is used as an interface for the amplifiers. To remove these lines, this control computer needs better electrical isolation between its digital and analogue sections. There are also peaks at 20Hz, which can be attributed to the air conditioning. The second amplifier has lower noise at high frequencies than the first amplifier since the DC gain of the amplifier is not the same as its frequency dependent gain.

The noise of the laser after the beam is transmitted through the PMC was measured (Figure 2.7). The PMC introduces length to intensity noise coupling via the offset in the PDH loop. This offset changes due to changing levels of RF coupling in the PMC servo electronics. As there was constant activity in the lab, this pick

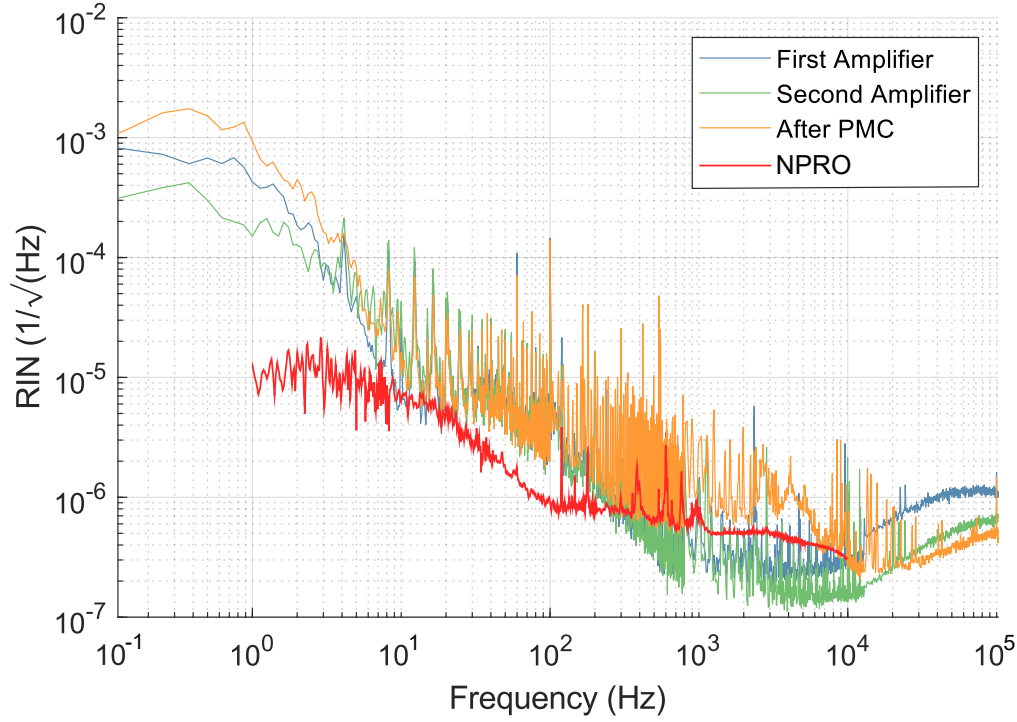


Figure 2.7

up changed - even moving a wire can cause a different amount of RF coupling. The coupling from length noise of the PMC to intensity noise as much as possible.

2.6.2 Actuator

The actuator that will be used to provide intensity stabilisation is an AOM. The use of a current shunt integrated into one of the pump diodes for Amp I was investigated as this has the potential to reduce the number of transmissive high power optics.

2.6.2.1 AOM

The AOM used is a Crystal Technology 3080-199. An AOM uses acoustic waves to scatter light into higher order modes. The amount of light that is Bragg scattered depends on the intensity of the acoustic wave. To control the intensity of the acoustic wave an RF driver, Gooch and Housego r21080-2am-rfx, was used. This driver produces an 80 MHz signal whose power can be modulated. By modulating the power of this signal, the power in the zeroth order beam can be controlled. The diffraction efficiency of this AOM was found to be 74%. This means that the AOM can be used for coarse and fine power adjustment. In order to modulate the intensity of a laser beam, some of it must always be scattered into the first order beam. In order to operate linearly, the DC level of the light that is dumped should be about 10W.

The transfer function between the amplifier power control of the AOM and the intensity of the zeroth order beam as measured by the ISS photodiode is shown in figure 2.9. The schematic for this measurement is shown in Figure 2.8. The signal needs to be linearised before going to the RF amplifier. The transfer function is shown in relative intensity units, i.e. the response was divided by the DC power on the photodiode. This transfer function is flat from 1Hz to 100kHz, and above 10kHz there is some phase lag.

2.6.2.2 Current Shunt

Modulating the current through a laser diode will modulate the intensity of the laser. This is a standard way of implementing intensity control at the 10W level [14, 15, 16, 17, 18]. Using a current shunt as an intensity actuator has been demonstrated

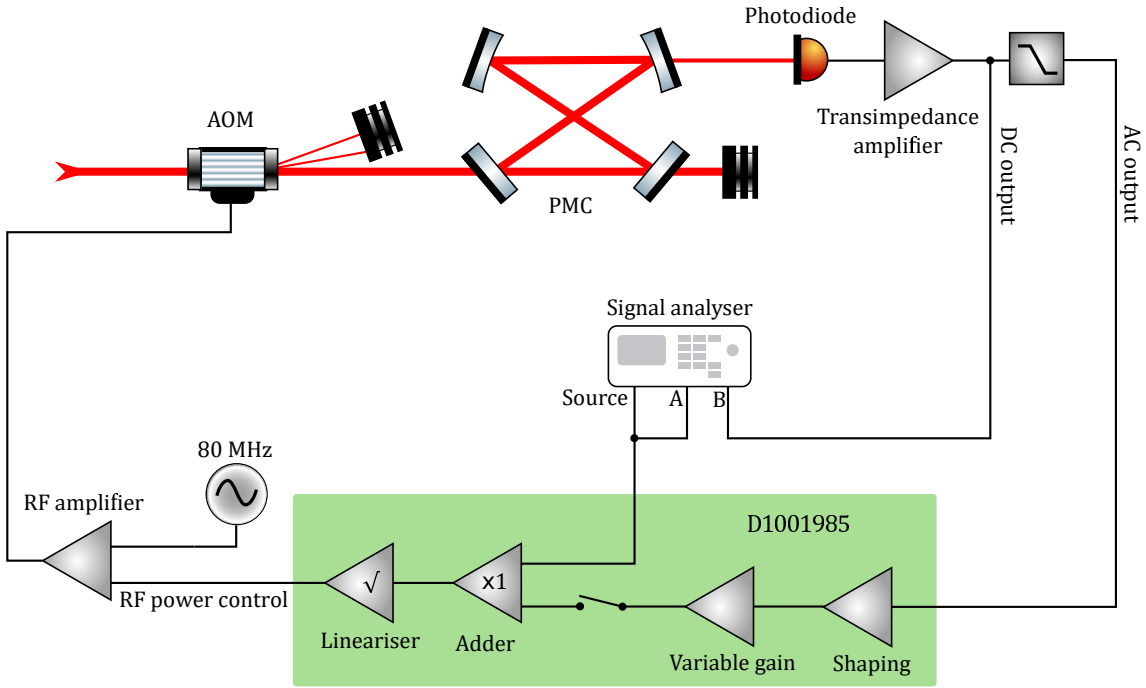


Figure 2.8

in a similar system at the 100W level [4].

The pump diodes current for the laser amplifier heads control the amount of power produced by the amplifier. By modulating the current through them, the power of the laser can be controlled. The current shunt diverts current from the pump diode by pulling it through resistor. This is controlled by a power transistor.

The current shunt technique allows for one to have fewer transmissive high power optics in the beam path, which potentially can reduce noise and thermal distortions of the beam. It requires a smaller DC offset compared to an AOM in order for it to work.

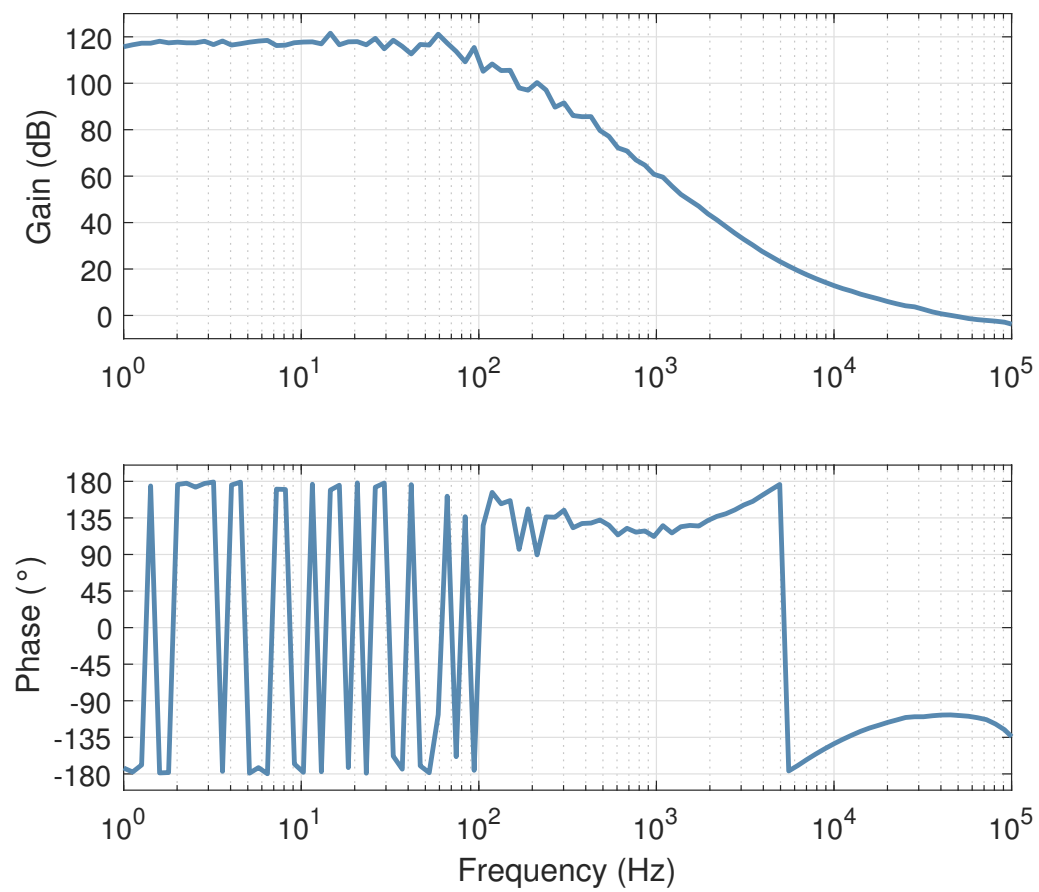


Figure 2.9

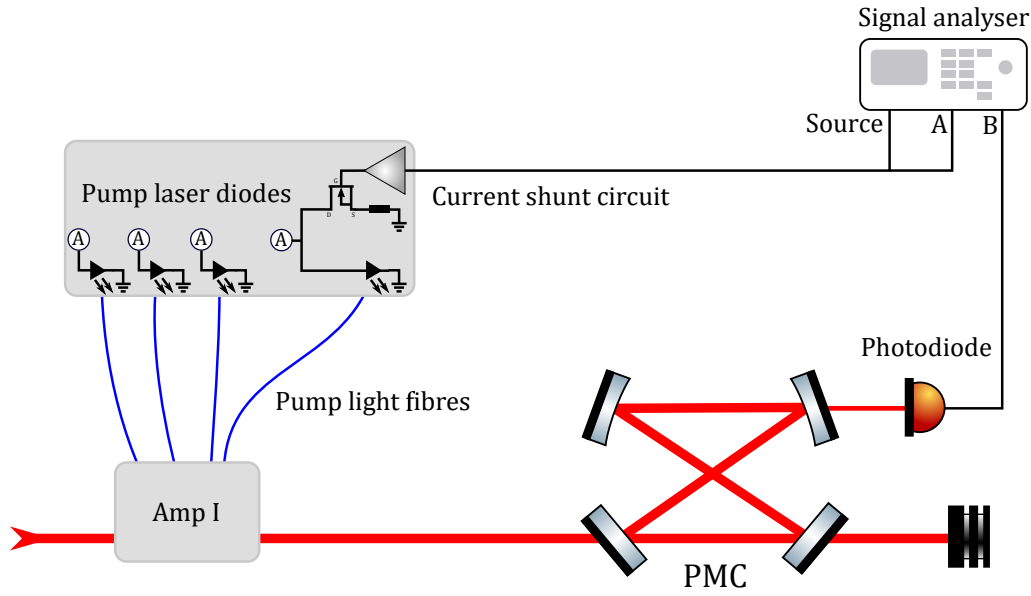


Figure 2.10

The transfer function between the diode current input and the ISS photodiode in transmission of the PMC was measured for a variety of different pump diode operating currents. A sketch of this is shown in figure 2.10. The transfer function was found to have significant amounts of variation in gain and phase between 1Hz-100kHz. It was found that the behaviour of the actuator at low frequencies strongly depended on the DC pump diode current. However, above 10kHz the transfer function was less dependant on the DC pump diode current. This is good, as the unity gain point of the intensity stabilisation servo will be between 10-100kHz, so the servo is more robust to changes in pump diode current. Figure 2.11 shows the transfer function, in relative intensity units, for the current shunt actuator with a variety of different current shunt offsets. The AOM is a much stronger actuator than the current shunt, by roughly 30 dB. This can be compensated for by changing the servo electronics.

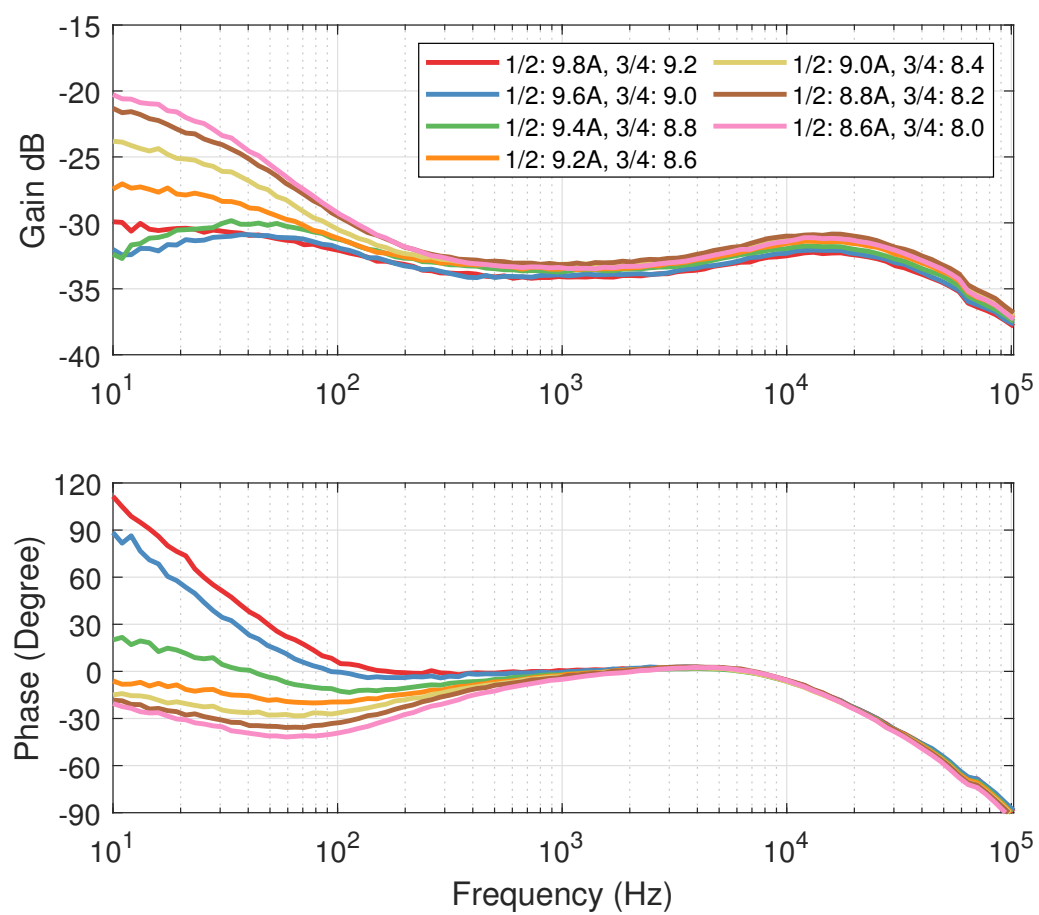


Figure 2.11

2.6.2.3 Reasoning for using the AOM over the Current Shunt

While the current shunt offers many advantages over the AOM, it has a major drawback: it cannot reduce the power at the many tens of Watts level. In order to do this, high power attenuation optics would be required, which means putting transmissive optics back in the beam bath and thus undoing the plus points gained by using a current shunt. When the LIGO detectors are able to operate at full power, then the current shunt actuator would be worth revisiting, however at present LIGO operates with between 25-50W of input power from the PSL.

2.6.3 Stabilised Noise Measurement

The loop is required to suppress the free running noise of the laser after the PMC, shown in Figure 2.7, to the inner loop requirement, shown in Figure 2.6. Therefore, the unity gain point of this servo must be above 10 kHz. At 10 Hz, the noise requirement is $2 \times 10^{-8} 1/\sqrt{\text{Hz}}$ and the noise is $2 \times 10^{-5} 1/\sqrt{\text{Hz}}$, so there needs to be at least 60 dB of open loop gain at this frequency.

The stabilised laser noise is shown in Figure 2.12. The out of loop noise was measured on the ‘spare’ ISS photodiode. A noise budget was created using the measured noise in the loop and the shot noise level of the light used to stabilise the laser. This agrees with the out of loop noise measurement to 10Hz. Below 10Hz, there is excess noise in the out of loop measurement compared to the noise budget. Possible sources of this noise could be scattered light or dust falling through the ISS beam. Near the unity gain point, due to the open loop transfer function phase and gain, there is a small increase in the stabilised noise compared to the free running

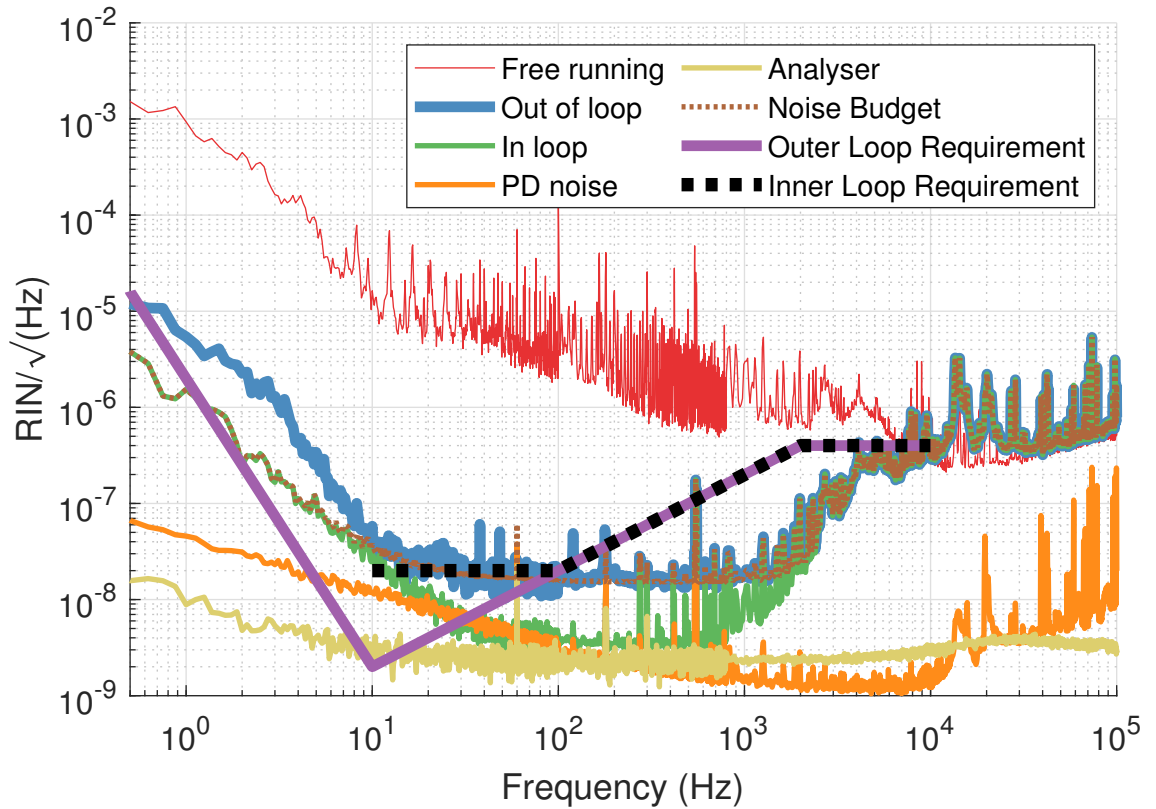


Figure 2.12

noise, however this is out of the requirement frequency band.

2.6.3.1 Open Loop Transfer function Measurement

A sketch of the servo is shown in Figure 2.13. The servo features two circuits that add shaping; these are found built into the photodiode circuit and within the rack electronics **why..?**. The rack electronics allow for the servo to be controlled via CDS. This gives the user the ability to turn the servo on/off, to vary the overall gain and activate integrators.

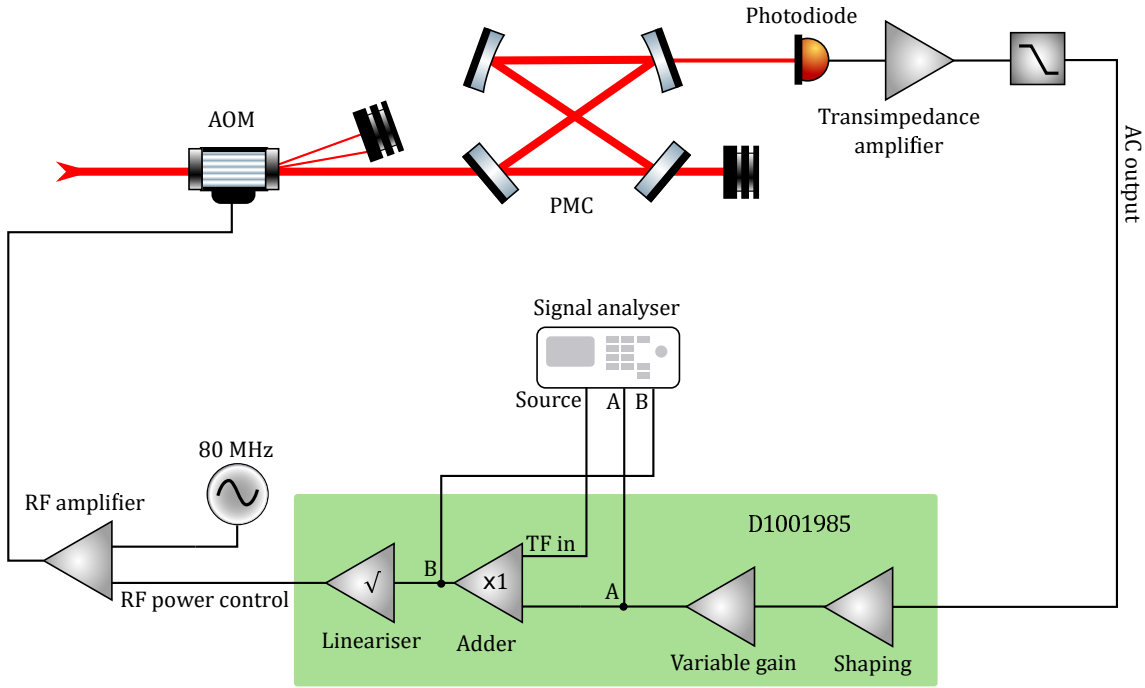


Figure 2.13

To measure the open loop transfer function, a swept sine into a times one adding stage of the servo. The signal at the other input to this stage was measured, and the signal immediately after this stage was measured. The ratio of these two gives the open loop transfer function of the servo. For this measurement, the gain in CDS was set to 20 dB. Once the intensity stabilisation was activated, the integrators were engaged to increase the open loop gain at low frequencies. The light transmitted through the PMC was 100 W. The open loop transfer function is shown in figure 2.14. The unity gain point is at 48 kHz with $\sim 70^\circ$ of phase margin. This servo satisfies the requirements.

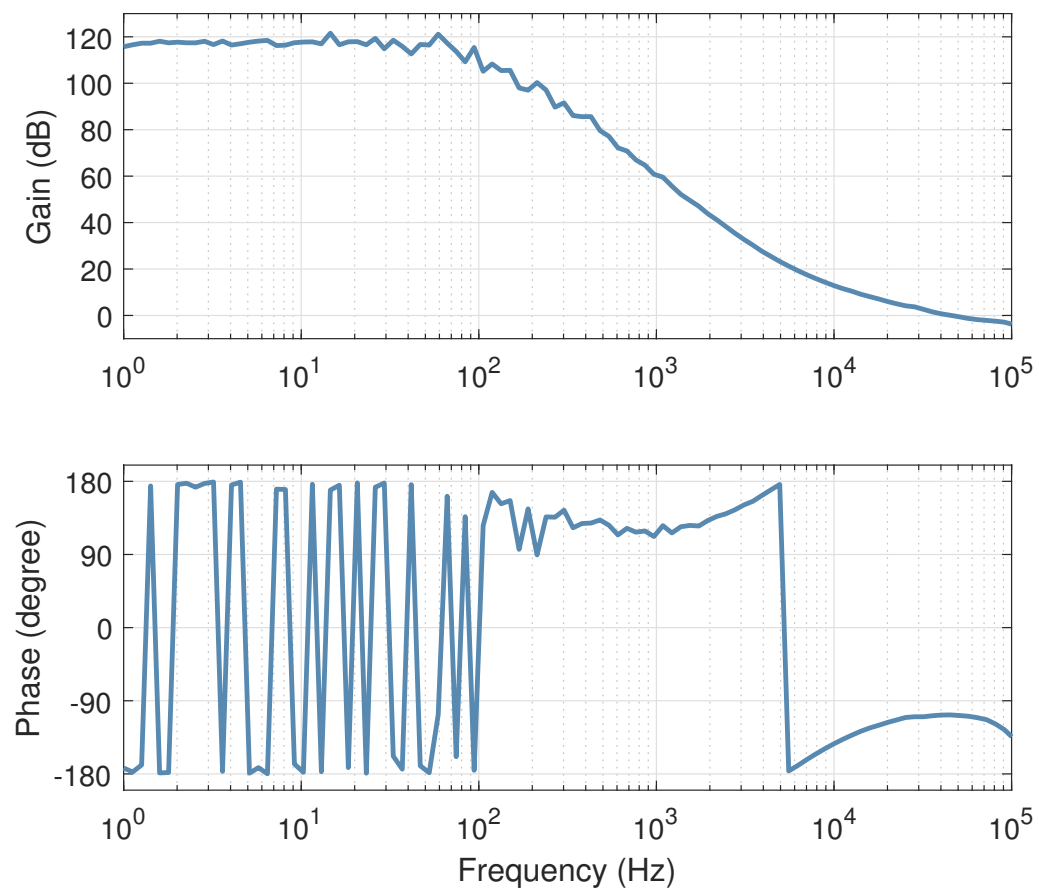


Figure 2.14

2.7 Conclusion

summary of how much power for how long, and how it could be improved slightly with mode matching

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